Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy

Abstract

Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers, intestinal debris—that correlate with the diagnosis; the standard defence restricts the model to the region of interest (ROI). This defence depends on where the artifact sits. In pre-registered traps built from real artifacts in four diseases and three modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (crossover +0.15 to +0.38 reversed-test AUROC, all Holm-significant; +0.25 on new patients). Moving the *same* real artifact across the ROI boundary of the *same* images reproduces the effect (location interaction hair +0.25, thyroid +0.28, ovary +0.07, capsule +0.50). It holds for five masking implementations, falls continuously with the artifact–ROI overlap, replicates across encoders, fine-tuning and zero-shot vision–language scores, and grows when images with contradicted labels are removed. A linear-Gaussian account predicted 84 results registered before they existed (mean absolute error 0.011 AUROC). On unaltered data the harm hides in subgroups: overall AUROC barely changes, but masking lowers AUROC on shortcut-conflicting cases, on a patient-disjoint thyroid split and on new patients, and lowers sensitivity among malignant nodules carrying a caliper for a near-benchmark fine-tuned model, in a subgroup registered in advance, partly even after re-tuning the threshold. Against a registered criterion, no remedy dominates masking; the closest, a class-conditional mean constraint on the masked features, removes masking's failure in every cohort and on every set of conflicting cases, at a cost of 0.02 overall AUROC on one external set. Audits of ROI-masked models should report artifact location and shortcut-conflicting performance. Code, pre-registrations and results are public.

Keywords: shortcut learning, spurious correlation, region-of-interest masking, artifacts, dermoscopy, ultrasound, capsule endoscopy

1. Introduction

Deep classifiers for medical images frequently rely on features that are correlated with the label in the training distribution but are not evidence of disease [10, 8]. In dermoscopy, surgical skin markings [38] and ruler scale bars [39] changed the melanoma predictions of a market-approved system; in chest radiography, pneumothorax classifiers detect the chest drain that treats the condition rather than the condition itself

[25, 29, 16, 41]. Such correlations are causal in origin—suspicious lesions are measured and marked, collapsed lungs are drained—so they are present in every retrospectively collected dataset.

The most widely recommended defence is to restrict the classifier to the region of interest (ROI): segment the lesion or the lungs and discard the rest of the image, or penalise what the model uses outside the segmentation [23, 1]. The rationale assumes that the shortcut lives in the background. Rulers, hair, calipers and debris do not respect that assumption. That a mask cannot remove what it keeps sounds self-evident; its consequences are not. We show that masking makes a model rely *more* on an artifact inside the ROI, that the resulting harm is hidden in overall metrics but concentrated in the cases a shortcut gets wrong, that it can be predicted before it is observed, and what to use instead. Three designs move the artifact relative to the ROI: controlled sweeps, traps built from real artifacts, and a transplant of real artifacts within the same images (Fig. 1).

Contributions.

1. **A location law, tested causally.** In traps built from real artifacts in four diseases and three imaging modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (Holm-significant crossovers in 4/4 cohorts). Moving the *same* real artifact across the ROI boundary of the *same* images reproduces the effect, so it is caused by location, not by which images carry artifacts. It holds for five masking implementations, is graded in the real overlap, replicates across encoders, end-to-end fine-tuning and new patients, and grows when images with contradicted labels are removed (Sec. 5).
2. **Mechanism and a closed-form account.** Masking retains an in-ROI artifact while removing competing evidence, so reliance on it grows. A two-parameter linear-Gaussian account states when in-ROI masking harms and predicted 84 results registered before they existed (Secs. 6, 7).
3. **Clinical consequences on unaltered data.** Masking leaves overall AUROC nearly unchanged but lowers it on shortcut-conflicting cases of a patient-disjoint thyroid split and of new patients; for a fine-tuned thyroid model near the published benchmark it lowers sensitivity among malignant nodules carrying a caliper, and part of that loss survives re-tuning the threshold on the test set (Sec. 8).
4. **What to use instead.** Against a registered criterion, tested in two rounds on newly generated data, no remedy dominates masking. The closest, a class-conditional mean constraint on the masked features, removes masking's in-ROI failure in every cohort and on every set of conflicting cases, at a small overall cost on one external set (Sec. 10).

Every confirmatory analysis was registered before its results, every reported interval is traced automatically to a result file, and code, registrations and result files are public.

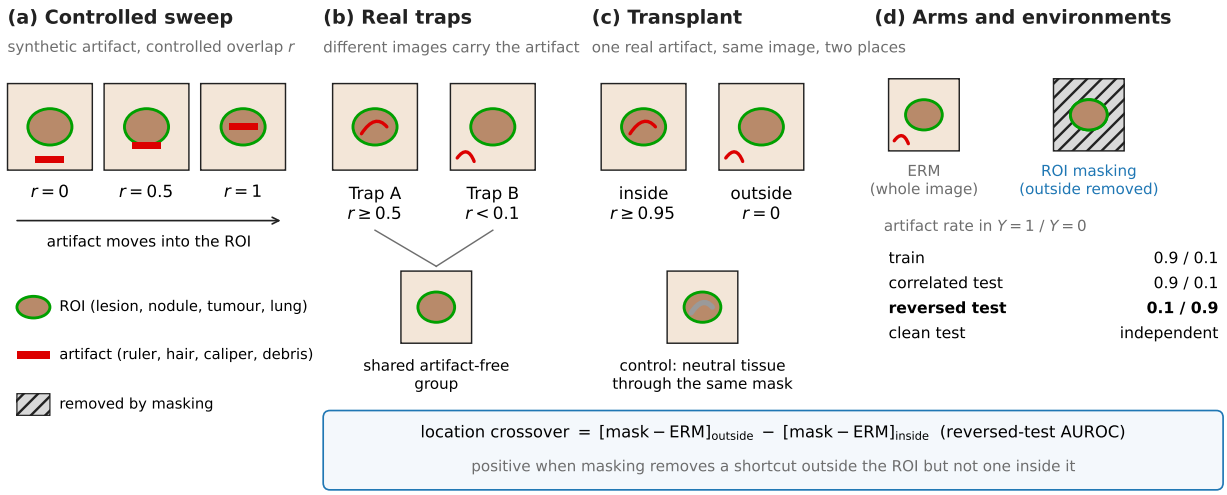


Figure 1: Study design. (a) Controlled sweeps paste a synthetic artifact at a controlled overlap r with the ROI. (b) Real traps compare images whose real artifact lies inside the ROI (Trap A) or outside it (Trap B), each added to one shared artifact-free group. (c) The real-artifact transplant pastes one real artifact instance inside or outside the ROI of the same artifact-free images, with neutral tissue pasted through the same mask as a paste-edge control. (d) ERM sees the whole image; ROI masking removes everything outside the ROI. Every design trains with the artifact on 90% of positive and 10% of negative images and tests on correlated, reversed and clean environments. Generated by `scripts/make_schematic.py`.

2. Related work

Artifacts as shortcuts. Bissoto et al. [4] annotated seven artifact types in ISIC images and showed that debiasing is harder than it looks; Nauta et al. [24] neutralised coloured calibration patches, which lie largely outside the lesion, by inpainting and retraining. Hair occlusion degrades classifiers trained on HAM10000 [36], and adding synthetic dark-corner artifacts helps more than removing real ones [27]; MaskMedPaint [17] inpaints regions outside the classification-relevant area. In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation [21]. In chest radiography, hospital signals limit generalisation [41], drain shortcuts are documented on NIH-CXR14 and CheXpert [25, 16], reduced by training with in-image annotations [29], localised to late layers of probed encoders [26] and mitigated by prevalence-equalised recalibration, which needs the artifact label at test time [18]. These works treat an artifact's presence. Location has been varied with a synthetic bias: placing it near to or far from the disease region in brain MRI changed performance gaps and how well mitigation worked [32]. Within our search, no study varies an artifact's position relative to the ROI used for masking, or does so with real artifacts.

Restricting the model to the ROI. Medical images often remain classifiable with the ROI removed, and models trained on full chest radiographs performed better on images without the ROI than on the ROI alone [31]. Training on segmented dermoscopy images improved balanced accuracy for one training set but not another [23], and penalising relevance outside segmentation masks makes chest-radiograph classifiers

more robust to background bias [1]. All of these assume that the shortcut lies outside the ROI. In natural images, mitigating one shortcut can amplify reliance on others [20]; masking is such a mitigation when an artifact lies inside the ROI, and we show that in medical images it amplifies the in-ROI shortcut, that location causes this, and why.

Removing shortcuts. Reweighting and GroupDRO [30], last-layer retraining [19] and linear concept erasure [28, 2] need artifact labels or paired data; erasure from unpaired labels removes label-correlated signal when the concept is correlated with the label. Task-preserving erasure (SPLINCE; 15) and projection of biased directions in vision–language models [6] need concept labels or prompts. Random erasing and occlusion augmentation [43, 9] use overlays as training noise, and test-time selection [3] steers dermoscopy models with user keypoints. Counterfactual invariance [34], enforced as conditional independence from an auxiliary label given the class [22], motivates the remedy we recommend.

Foundation models. Classifiers trained on frozen foundation-model features inherit shortcut-like behaviour that pooled multi-institutional evaluation can hide [11], and shortcut testing [5] probes whether performance gaps stem from encoding a sensitive attribute. We ask where in the image the shortcut sits and what masking can and cannot remove.

3. Data

Dermoscopy. Controlled cohort: ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels (ISIC 2016/2017 tables); 2,437 images on which a 65×19 px ruler can be placed at 0/25/50/75/100% overlap with the lesion. *Real hair:* all 25,331 ISIC 2019 training images; hair masks from Wegley et al. [37]; lesion masks manual for HAM10000 [33] and otherwise from a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels): Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma), subsampled to the hair-free source mix within each label; five seeds of five-fold cross-validation grouped by lesion identifier and perceptual hash. *New patients:* ISIC 2020 (32,997 images of 2,056 patients), hair masks from a segmenter trained only on ISIC 2019.

Thyroid ultrasound. TN3K images [12] with benign/malignant labels from cytological biopsy [13] (3,493 images; 1,210 malignant) and manual nodule masks; the official train/test split (2,879/614 images) is patient-disjoint. Sonographer calipers are segmented by a frozen rule-based detector (dotted measurement lines and '+' ends). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801.

Ovarian tumour ultrasound. MMOTU OTU_2d [42] (CC BY 4.0): 1,202 images with tumour masks, benign cystic versus lesions with solid components; the thyroid caliper detector is applied unchanged (Trap A 341/276, Trap B 91/103, caliper-free 170/178).

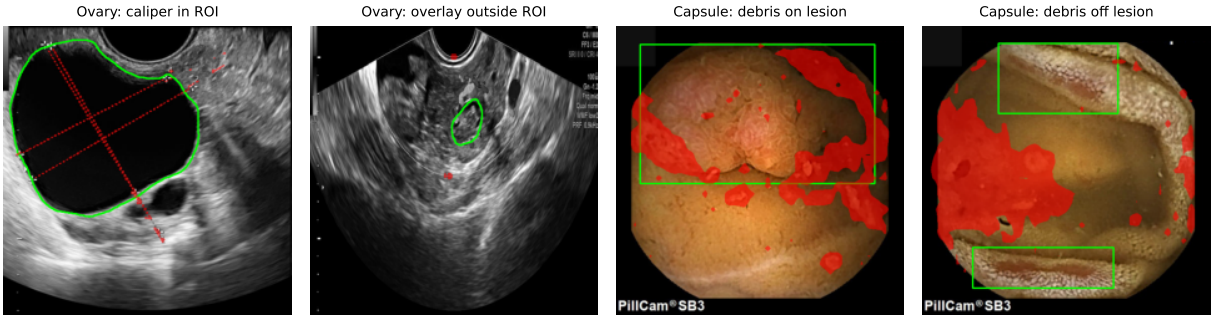


Figure 2: Real in-ROI and out-of-ROI artifacts (openly licensed cohorts). Green: ROI; red: artifact mask. Left: MMOTU ovarian tumours (caliper inside the tumour; overlay outside). Right: SEE-AI capsule frames (debris on and off the lesion box).

Capsule endoscopy. SEE-AI frames [40] (CC BY 4.0) with only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Debris, bubbles and bile are segmented by a linear probe on DINOv2 patch tokens (held-out IoU 0.62 against expert masks). Leakage groups: blocks of 100 consecutive frames plus perceptual-hash duplicates.

Chest radiography. NIH ChestX-ray14 [35] with lung ROIs from the TorchXRayVision PSPNet [7]: a controlled tube sweep on 7,610 images, 29,687 images with real drains, and devices from RANZCR-CLiP linked to NIH (Supplement S3).

4. Problem setup and protocol

Let $Y \in \{0, 1\}$ be the diagnosis and $A \in \{0, 1\}$ the presence of an artifact. A *trap* trains with $P(A{=}1 \mid Y{=}1) = 0.9$ and $P(A{=}1 \mid Y{=}0) = 0.1$ and tests on a correlated environment with the same rates, a *reversed* environment with the rates swapped, and a clean environment in which the artifact is independent of the label. A model that uses A gains on the correlated test and collapses on the reversed one; the primary metric is reversed-test AUROC, and clean AUROC measures the cost of a mitigation. The overlap $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$ is the variable of interest. Three designs vary it: *controlled sweeps* paste a synthetic artifact at overlaps 0–100% on artifact-free images; *real traps* compare images whose real artifact lies inside the ROI ($r \geq 0.5$, Trap A) or outside it ($r < 0.1$, Trap B), each added to one shared artifact-free group; the *real-artifact transplant* pastes one real artifact instance inside ($r \geq 0.95$) or outside ($r = 0$) the ROI of the same artifact-free images. The *location crossover* $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC summarises each design (Fig. 1). Unless stated otherwise backbones are frozen and linear heads are fitted; regularisation strength and the decision threshold (balanced accuracy) are selected on clean validation data only. Splits are grouped by lesion or patient identifier where published and otherwise by perceptual hash (and frame blocks).

Statistics. Uncertainty is a paired *crossed* bootstrap: training seeds (or cross-validation folds) are resampled as clusters, each test image receives one Poisson weight shared by every cluster in which it appears, and AUROC differences are computed inside each cluster (10,000 replicates unless stated). *Every interval of a model comparison in the paper and the supplement is a crossed-bootstrap interval*, computed from regenerated per-image predictions; the per-seed bootstrap of the original protocol, which treats a test set shared by the seeds as several and gave intervals a median 1.56 times narrower, is not used (Supplement S15). Agreement statistics between label sources, which involve no training seeds, use an image bootstrap, and seven secondary analyses in the supplement come from archived runs without per-image predictions and are reported as point estimates, marked as archived. Each family of tests is Holm-corrected.

Registration. Every confirmatory analysis was registered in a document committed before its results; the registry with commit hashes and times is in Supplement S1. The real traps, sweeps and unaltered-data tests were registered first (the 16-test primary family was grouped afterwards); the matched traps, transplant, paste-edge control and operating points were designed after the real-trap results and registered before their own results; each later registration was pushed to GitHub before its analysis was run. Analyses decided after their data were seen are labelled exploratory.

5. Results: a location law for ROI masking

5.1. Controlled sweeps

With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14, three seeds), ROI masking raises reversed-test AUROC over ERM by $+0.184$ when the ruler lies outside the lesion and lowers it by -0.129 when the ruler lies on it (location interaction $+0.313$ $[+0.258, +0.370]$). At full overlap the masked model's correlated AUROC exceeds ERM's (0.961 vs 0.937): it uses the ruler *more*. Synthetic calipers on thyroid nodules and ovarian tumours and synthetic debris on capsule lesions show the same pattern (Fig. 3; all intervals in Supplement S14).

5.2. Real artifacts in four diseases (pre-registered)

In the real traps the crossover $[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ is positive and Holm-significant in all four cohorts, from $+0.147$ for dermoscopic hair to $+0.377$ for capsule debris (Table 1, first column). For hair, masking *harms* when the hair lies on the lesion (-0.049 $[-0.083, -0.018]$) and helps when it lies beside it ($+0.098$). For calipers and debris, in-ROI masking still helps, but far less than out of the ROI (thyroid $+0.109$ versus $+0.348$), and the masked Trap A model still relies on the calipers (correlated 0.908 versus reversed 0.398). The registered external test on new patients (ISIC 2020, patient-disjoint folds, hair masks from a segmenter frozen before any ISIC 2020 image was scored) reproduces the hair pattern, with harm on the lesion (-0.064), gain beside it ($+0.186$) and a crossover of $+0.250$ $[+0.174, +0.327]$. What is universal

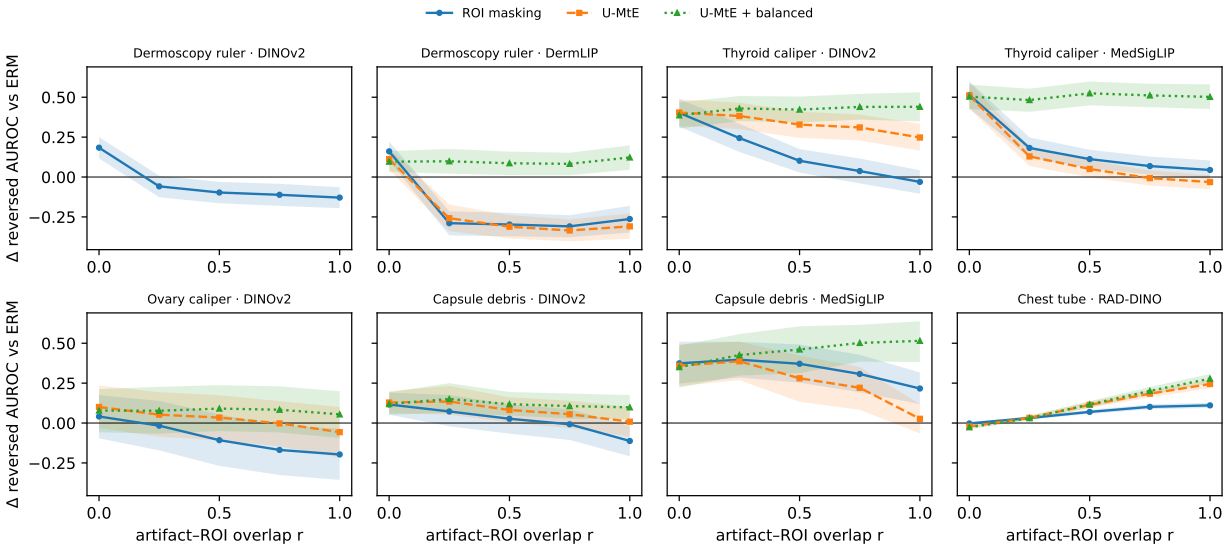


Figure 3: Controlled overlap sweeps: change in reversed-test AUROC versus ERM as the synthetic artifact moves into the ROI (95% CIs shaded). ROI masking (blue) declines with overlap in every modality except chest radiography, where the encoder ignores extra-pulmonary tubes; U-MtE with group balancing (green) is flat. Generated by `scripts/make_figures.py`.

is therefore the *gap*: masking leaves an in-ROI shortcut largely intact while it removes an out-of-ROI one. Harm is conditional; it appears when the mask also removes disease context or leaves the artifact a larger share of what remains (Sec. 7).

5.3. Location, not image population: transplant and matched traps

Trap A and Trap B contain different images, and these differ systematically: lesions with hair on them are much larger (standardised mean difference 2.1), as are thyroid nodules with calipers on them and capsule lesions with debris on them. We therefore cut real artifact instances—a caliper line with its end crosses, a hair strand, a piece of debris—from artifact-bearing images and pasted each *inside* or *outside* the ROI of the *same* artifact-free images, with the same instance, transform and compositing at both locations (pre-registered). Outside the ROI masking removes the pasted shortcut entirely; inside, the shortcut survives, and the location interaction is positive in all four cohorts (Table 1, third column). A paste-edge control, neutral tissue pasted through the same masks, gives 52–92% of the effect: what the transplant isolates is the location of a label-correlated pattern, not the appearance of a particular artifact; the artifact's own content adds to it significantly for hair and thyroid calipers only (last column). As an observational check we matched the artifact-bearing images of the two traps 1:1 on lesion size, artifact amount and acquisition covariates. The crossovers are unchanged (second column), shrink by at most 16% under a four times stricter calliper, and survive a doubly robust regression on every matching covariate (Holm $p = 0.002$ in each cohort). Matching was infeasible for capsule (19 pairs), where the location evidence rests on the transplant. Details

Table 1: The location law under four designs (reversed-test AUROC, DINOv2 ViT-B/14; 95% crossed seed×image bootstrap CIs, Sec. 4). *Real traps*: crossover $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ between different artifact-bearing images (rebuilt run). *Matched*: the same after 1:1 propensity matching of the in-ROI and out-of-ROI artifact-bearing images (calliper 0.2 SD; Supplement S11). *Transplant*: one real artifact instance pasted inside or outside the ROI of the same artifact-free images (location interaction). *Neutral paste*: neutral tissue pasted through the same masks, positions and blending (paste-edge control). Transplant and matching were designed after the real-trap results and registered before their own results; each design is its own Holm family.

Cohort	Real traps	Matched	Transplant (artifact)	Neutral paste	Artifact − neutral
ISIC hair	$+0.147$ $[+0.109, +0.185]$	$+0.154$ $[+0.122, +0.187]$	$+0.253$ $[+0.214, +0.293]$	$+0.204$ $[+0.164, +0.244]$	$+0.049$ $[+0.026, +0.072]$
Thyroid	$+0.239$ $[+0.196, +0.280]$	$+0.238$ $[+0.186, +0.291]$	$+0.277$ $[+0.256, +0.298]$	$+0.145$ $[+0.130, +0.161]$	$+0.132$ $[+0.116, +0.148]$
Ovary	$+0.170$ $[+0.087, +0.253]$	$+0.163$ $[+0.049, +0.274]$	$+0.074$ $[+0.044, +0.104]$	$+0.052$ $[+0.021, +0.084]$	$+0.022$ $[-0.002, +0.048]$
Capsule	$+0.377$ $[+0.326, +0.425]^{\ddagger}$	infeasible†	$+0.497$ $[+0.427, +0.567]$	$+0.459$ $[+0.383, +0.535]$	$+0.037$ $[-0.017, +0.093]$

†19 matched pairs in one label, below the pre-registered minimum of 30. ‡Descriptive: capsule lesions carrying debris are much larger than those with debris beside them (standardised mean difference 2.6) and matching is infeasible, so this crossover is not by itself evidence about location.

and balance tables: Supplement S11.

5.4. How general the law is

All four other masking implementations—black fill, blurred background, crop to the lesion box, crop of the masked image—give a positive crossover in all four cohorts (16 of 16, Holm $p \leq 0.010$). The two traps are the ends of a continuous dependence: binned by their real overlap, the mask gain falls with r in every cohort, and for hair it turns from a gain into a harm near $r = 0.4$. The crossover replicates with MedSigLIP in three cohorts, with a CNN (ConvNeXt-Base) in two, with DINOv2 at three sizes (nine of nine), with end-to-end fine-tuning in the three cohorts where the network learned the task, and in the zero-shot scores of vision–language models. The one exception is the dermatology encoder DermLIP: masking costs it disease signal at both hair locations, and the hair crossover vanishes ($+0.014$ $[-0.037, +0.063]$), as the account of Sec. 7 anticipates. Details: Supplement S7, S16 and S18.

5.5. Chest radiography: a boundary case

A radiograph encoder (RAD-DINO) ignores a synthetic tube outside the lungs, so masking has nothing to remove there. On real devices (RANZCR-CLiP linked to NIH), lung masking helped at both locations with RAD-DINO; with MedSigLIP, which does use the out-of-lung tube, the crossover is positive for all four findings. The device contrast is, however, confounded with other tubes, and real chest drains carry the shortcut through the treated lung. We report chest radiography as the boundary of the law, not as a fifth confirmation (Supplement S3).

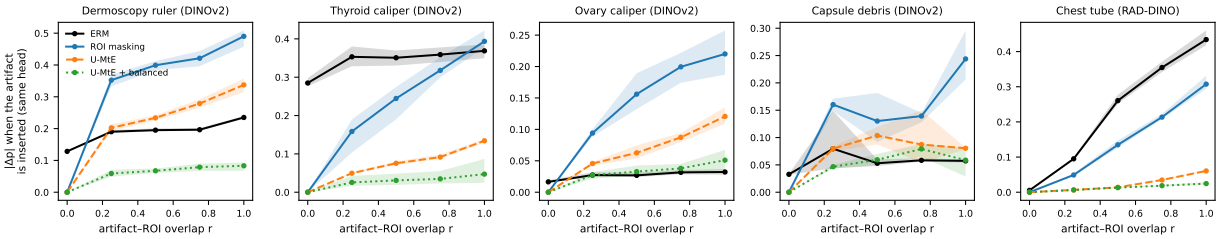


Figure 4: Mechanism: same-head counterfactual sensitivity $|p(x{+}\text{artifact}) - p(x)|$ versus the artifact's overlap with the ROI in the five controlled sweeps (mean over seeds; band: range over seeds). Masking removes the out-of-ROI artifact ($r = 0$) but increasingly relies on an in-ROI one; erasure and balancing remove that reliance. Generated by `scripts/analysis/mechanism_figure.py`.

6. Mechanism: masking retains an in-ROI artifact and removes its competitors

Three explanations were tested in the controlled ruler sweep. *Occlusion* alone does not harm: with the ruler present but uncorrelated with the label, masking changes AUROC by at most +0.041 at any overlap, never significantly, so the harm requires the correlation. *Lost context* is not the driver: widening the mask so that it restores ruler pixels increases the harm. *Amplified reliance* is: with the head held fixed, inserting the ruler on the lesion changes the masked model's prediction about twice as much as ERM's ($|\Delta p|$ 0.491 versus 0.235 at full overlap; paired difference $+0.255$ $[+0.221, +0.287]$), because masking removes the competing evidence around it. The same holds for ovarian calipers and capsule debris; for thyroid calipers the masked model relies on the caliper as much as ERM does (Fig. 4). Harm is largest for small lesions, where the ruler is a larger share of what survives the mask (Supplement S14).

For real hair, removing every hair pixel at test time closes only 44–47% of ERM's clean–reversed gap: most of what a frozen encoder exploits is what accompanies hair (age, sex, body site), not the hair itself. This bounds every pixel-level method and explains why methods that use image-level artifact labels, which remove everything correlated with the label, recover more. It also limits the transplant: a pasted hair carries the shortcut in its pixels by construction, so the transplant shows the location effect for pixel-carried hair, not its size for hair in the wild.

7. Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A{=}1 \mid Y{=}1) = p_1$, $P(A{=}1 \mid Y{=}0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact's effective training SNR is

$\tilde{A} = (p_1 - p_0)^2 A/(1 + p_1(1 - p_1)A)$ and

$$\mathrm{AUROC}_{\mathrm{rev}}(S, \tilde{A}) = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \tag{1}$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Supplement S14). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$: out of the ROI $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) in-ROI masking harms when the mask removes disease-informative context ($S_m < S$), and can harm even when it does not if the artifact becomes a larger share of what remains ($\tilde{A}_m > \tilde{A}$); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \to 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure fails when the artifact resembles the pathology unless the disease direction is protected; (iv) group balancing removes the artifact–label association and everything correlated with it, whatever the location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections 5–10. In the terms of Hermann et al. [14], who separate a feature's predictivity from its availability, masking changes the relative predictivity of what the head can still use; unlike that synthetic analysis, the account is fitted to trained models and makes quantitative predictions.

Does the account match real encoders?. The same model gives the AUROC for any test composition, so (S, A) can be fitted to each trained model's clean and correlated AUROC *only* and the reversed-test AUROC predicted from them; no parameter is shared across models or tuned. Table 2 compares its error with two reference predictors that use the same two inputs, a linear-symmetry heuristic (reversed = 2·clean−correlated) and "reversed = clean" (Fig. 5). On the results that existed when the test was designed the account's error is less than half the heuristic's, and it anticipates DermLIP's missing hair gap; for the crossover itself the heuristic does almost as well, so we do not use the crossover correlation as evidence. With the chest-radiograph cells added, one near-zero device crossover has the wrong sign, which fails our pre-registered all-signs criterion, and the account fails, as consequence (v) says it must, for chest drains, where the shortcut is carried by the treated lung (implied reversed AUROC 0.72–0.78, observed 0.19–0.23). Across all cohorts it is therefore a qualitative account.

Prospective test (pre-registered). We then registered the same procedure, unchanged, before the masking-implementation and overlap-bin analyses were run. On their 84 cells the account's error was 0.011 AUROC, a third of the heuristic's, with every crossover sign right (Table 2), which meets its registered criterion for quantitative prediction within frozen-encoder linear heads on the four real-artifact cohorts.

Table 2: Held-out reversed-test AUROC predicted from each model's clean and correlated AUROC only: mean absolute error of the linear-Gaussian account and of two reference predictors that use the same inputs, the account's correlation with the observations, and the crossovers whose sign it gets right.

Test	Models	Account	Symmetry heuristic	Reversed = clean	r	Crossover signs
Retrospective: real traps and controlled sweeps	136	0.052	0.129	0.237	0.91	9/9
with the chest-radiograph cells added	154	0.053	0.125	0.245	0.88	12/13
Prospective, registered before the data existed	84	0.011	0.036	0.177	0.996	16/16

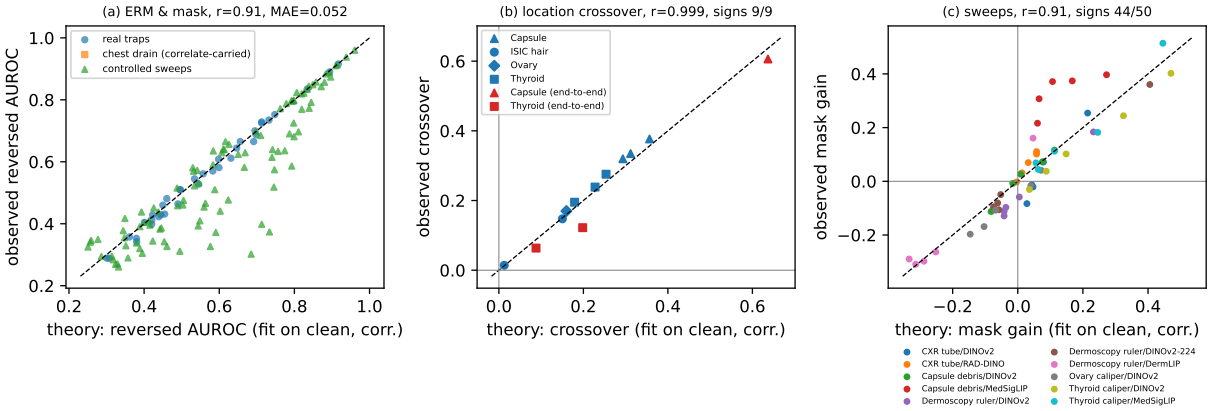


Figure 5: Pre-registered test of the linear-Gaussian account on real encoders. (S, A) are fitted to each model's clean and correlated AUROC; the reversed-test results are held out. (a) Implied vs observed reversed-test AUROC (ERM and masking). (b) Location crossover per cohort and backbone; red: end-to-end fine-tuned networks, outside the model's assumptions. (c) Mask − ERM gain along the controlled overlap sweeps. Dashed: identity. Generated by `scripts/analysis/theory_predict.py`.

8. Consequences on unaltered test data

The traps construct an extreme reversal. Without any resampling, in-ROI artifacts are still associated with the label (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural training data and tested on untouched data: the official, patient-disjoint TN3K/TNCD test split [13], each ISIC source held out in turn, and held-out capsule frame blocks. We report AUROC on *shortcut-conflicting* pairs, in which the artifact points to the wrong diagnosis, beside AUROC over all pairs.

Discrimination. Table 3 collects every unaltered-data result. On shortcut-conflicting pairs masking *lowers* AUROC in three of five test sets (thyroid -0.095 $[-0.167, -0.023]$ and two of the three held-out dermoscopy sources), raises it for the third and leaves capsule unchanged. Over all thyroid pairs it changes AUROC by only -0.004: the harm is confined to the cases in which the in-ROI artifact points to the wrong diagnosis,

Table 3: Masking on the unaltered test data: mask − ERM with crossed 95% CIs (mean over five seeds). *Conflicting pairs*: positive–negative pairs in which the in-ROI artifact points to the wrong diagnosis. Sensitivities at thresholds fixed on validation data, except where re-tuned on the test set so that both arms reach specificity 0.80. The 78 conflicting nodules are the malignant nodules of the test split that carry an in-ROI caliper; for the frozen model they were defined after its first unaltered-data analysis, for the fine-tuned model before any of its results. All five operating points: Supplement S9.

Measure	ERM → mask	mask − ERM [95% CI]	Status
Thyroid, official patient-disjoint split: frozen DINOv2			
AUROC, all pairs		-0.004 $[-0.048, +0.041]$	registered
AUROC, conflicting pairs		-0.095 $[-0.167, -0.023]$	registered
Sensitivity at validation specificity 0.80	0.557 → 0.355	-0.202 $[-0.300, -0.100]$	registered
78 conflicting malignant nodules	0.549 → 0.177	-0.372 $[-0.499, -0.236]$	registered test, post hoc subgroup
same, threshold re-tuned on the test set	0.505 → 0.331	-0.174 $[-0.313, -0.012]$	registered
Same split: fine-tuned ConvNeXt-T (test AUROC 0.769)			
AUROC, conflicting pairs		-0.102 $[-0.178, -0.026]$	registered
Sensitivity at validation specificity 0.80	0.747 → 0.371	-0.376 $[-0.462, -0.297]$	registered
78 conflicting malignant nodules	0.744 → 0.223	-0.521 $[-0.642, -0.382]$	registered
same, threshold re-tuned on the test set	0.544 → 0.390	-0.154 $[-0.302, -0.023]$	registered
Dermoscopy, one ISIC 2019 source held out: frozen DINOv2			
AUROC, conflicting pairs, BCN held out		-0.025 $[-0.042, -0.008]$	registered
AUROC, conflicting pairs, HAM held out		$+0.050$ $[+0.030, +0.070]$	registered
AUROC, conflicting pairs, MSK held out		-0.065 $[-0.111, -0.020]$	registered
New patients: trained on ISIC 2019, tested on ISIC 2020			
AUROC, all pairs		$+0.016$ $[-0.000, +0.033]$	sensitivity analysis
AUROC, conflicting pairs (label-free duplicate rule)		-0.037 $[-0.061, -0.012]$	sensitivity analysis
AUROC, conflicting pairs (registered duplicate rule)		-0.026 $[-0.052, -0.000]$	registered
Capsule endoscopy, held-out frame blocks			
AUROC, all pairs		$+0.077$ $[+0.059, +0.097]$	registered
AUROC, conflicting pairs		$+0.021$ $[-0.003, +0.048]$	registered

which is exactly where a shortcut is dangerous.

Operating points (pre-registered). A clinical decision uses a threshold. With thresholds fixed on each seed's validation split, masking lowers sensitivity on the official thyroid split at all five operating points (Supplement S9). Specificity rises correspondingly, so part of the change is a shift of the scores relative to the validation thresholds. The loss is largest where the shortcut conflicts with the label: among the 78 malignant nodules that carry an in-ROI caliper, sensitivity at a validation specificity of 0.80 falls from 0.549 to 0.177. For this frozen model the subgroup was defined after its AUROC harm was found and only the operating-point test was registered in advance.

A fine-tuned model near the benchmark (pre-registered). To test whether these are properties of a weak model, we chose a fine-tuning recipe on validation images only (ConvNeXt-T pretrained on ImageNet-22k), registered the subgroup test before any result of this model existed and trained it with five seeds. Its test AUROC is 0.769, 0.004 below the lowest published cross-entropy baseline on this split (0.773; 13). All four registered tests are supported (Holm): the loss on conflicting pairs, the loss of sensitivity at four of five operating points, the loss among the 78 conflicting nodules—from 0.744 to 0.223 (-0.521 $[-0.642, -0.382]$)—and a positive trap crossover.

Is the loss a threshold artefact? (pre-registered). Re-tuning each model's threshold on the test set itself—the best that recalibration at a deployment site can achieve—makes the overall sensitivity of the two arms equal, but the conflicting subgroup still loses, for the fine-tuned model (-0.154 $[-0.302, -0.023]$) and for the frozen probe (Holm $p = 0.019$ each; Table 3). About 30% of the loss at validation-fixed thresholds therefore cannot be removed by any threshold; the rest follows from transporting thresholds from validation to test data, as deployment does.

New patients at natural prevalence (pre-registered). Trained on all ISIC 2019 images and tested on ISIC 2020 without resampling, masking lowers the AUROC on conflicting pairs under both duplicate rules, while overall AUROC is nearly unchanged; the registered rule proved too loose (it removed most look-alike images of other patients), so a label-free rule was registered as a sensitivity analysis. Group balancing recovers the conflicting pairs, and no sensitivity loss at a fixed operating point was detectable (Supplement S18).

9. Robustness of the evidence

Artifact labels, checked without a clinician (pre-registered). The labels come from automatic detectors. Stricter label definitions (large markers only; stricter location thresholds) leave every thyroid and ovary crossover significant and larger, as expected if label errors dilute rather than create the effect. We also compared the labels with every independent source we could find. Where hair is present our labels place it

correctly (lesion masks against multi-annotator masks: Dice 0.851), but our hair-presence threshold marks hair on many images that expert-reviewed labels call hair-free ($\kappa = 0.076$); removing every image an independent source contradicts *raises* the hair crossover from +0.147 to +0.199 [+0.164, +0.233]: the label noise diluted the effect. For thyroid calipers two independent labellers (an ultrasound annotation detector and a locally run medical vision–language model) agree with our cell for 94% of the images on which they agree with each other; the registered cleaning rule, which removes an image whenever either disagrees, left too few out-of-ROI images to be tested; without the images both contradict, the crossover is +0.257 (exploratory; +0.239 on our labels). The ovary and capsule labels could not be checked with public data.

Leakage and rebuild. Uniting the perceptual-hash groups with connected components of DINOv2 embeddings (pre-registered) keeps 11 of 12 primary contrasts at the same sign and significance. A complete rebuild on different hardware, with every derived object retrained or recomputed, reproduces every primary crossover within 0.01. Details: Supplement S18.

10. What to use instead of masking

A class-conditional constraint on the masked features. Let x be the frozen features of the masked image, y the diagnosis and a the image-level artifact label from the automatic detectors of Sec. 3, used during training only. On the training images we compute the within-class mean differences $\delta_y = \mu(x \mid a{=}1, y) - \mu(x \mid a{=}0, y)$, project the features onto the orthogonal complement of δ_0, δ_1 (or of their average; chosen on validation data) and fit an ordinary logistic head on all training images at their natural weights. Within each class, the head then cannot score images with and without the artifact differently on average, and no image is down-weighted; group balancing, by contrast, loses on unaltered data mainly by discarding disease evidence through reweighting (Supplement S17). The constraint is the first-moment, linear case of counterfactual invariance [34] and of the auxiliary-label conditional-independence regularisers of Makar et al. [22]; what is new is its derivation from the location law and its use after masking. At test time it needs only the masked image.

Registered test against masking. We registered a criterion for *dominating* masking: not worse (margin 0.01 AUROC) on the six unaltered test sets and on the Trap B and clean tests, better on every Trap A and on the conflicting pairs of the unaltered test sets, and no shortcut flipping—42 components, all of which must hold (Supplement S17). Settings were chosen on validation data only; confirmation used trap resamples generated after the registration and one cohort (ovary) that never influenced a choice. Table 4 compares every remedy with masking on these data. Group balancing, alone or after U-MtE, gains the most where masking fails, but pays where masking already works: it is worse than masking on the unaltered test sets overall in three of six (e.g. ISIC 2020 −0.025 and −0.035), because reweighting discards disease evidence.

Table 4: AUROC of every method on the confirmation data of the first registered round (new trap resamples and the unaltered test sets; frozen DINOv2). Traps: the worse of reversed and correlated AUROC, so a method cannot score by flipping the shortcut. ↑/↓: better/worse than masking, 95% crossed CI of the difference entirely above/below zero (all intervals in Supplement S17). *Ovary never influenced a choice. The class-conditional constraint was the registered candidate: the criterion allowed a loss of at most 0.01 on all pairs and required gains in every Trap A and on every set of conflicting pairs, and it met 40 of its 42 components; the other columns are comparators fitted on the same data.

Test	ERM (no mask)	ROI masking	Group balancing	U-MtE	U-MtE + balancing	Class-conditional constraint
Needs artifact labels	no	no	yes	no	yes	yes
Trap A: artifact inside the ROI						
Thyroid	0.289↓	0.400	0.670↑	0.624↑	0.793↑	0.730↑
Capsule	0.587↓	0.667	0.767↑	0.548↓	0.855↑	0.820↑
Dermoscopy hair	0.506↑	0.451	0.738↑	0.492↑	0.679↑	0.644↑
Ovary*	0.431↓	0.520	0.698↑	0.588↑	0.733↑	0.638↑
Trap B: artifact outside the ROI						
Thyroid	0.408↓	0.744	0.635↓	0.744	0.770	0.759
Capsule	0.414↓	0.882	0.658↓	0.862↓	0.886	0.903↑
Dermoscopy hair	0.522↓	0.636	0.734↑	0.646	0.756↑	0.681↑
Ovary*	0.442↓	0.716	0.679	0.737	0.753	0.714
Unaltered test sets, all pairs						
Thyroid (official split)	0.739	0.731	0.721	0.740	0.732	0.729
Capsule	0.907↓	0.972	0.904↓	0.965↓	0.963↓	0.972
ISIC, BCN held out	0.786↑	0.759	0.782↑	0.765↑	0.769↑	0.755↓
ISIC, HAM held out	0.754↓	0.791	0.750↓	0.801↑	0.807↑	0.788
ISIC, MSK held out	0.768	0.782	0.765	0.776	0.769↓	0.780
ISIC 2019 → 2020	0.795	0.806	0.781↓	0.791↓	0.772↓	0.785↓
Unaltered test sets, conflicting pairs						
Thyroid (official split)	0.670↑	0.578	0.718↑	0.613↑	0.653↑	0.663↑
ISIC, BCN held out	0.719↑	0.685	0.739↑	0.688	0.723↑	0.721↑
ISIC, MSK held out	0.688↑	0.623	0.725↑	0.614	0.652↑	0.694↑
ISIC 2019 → 2020	0.786↑	0.731	0.818↑	0.734	0.774↑	0.809↑

The constraint keeps most of that gain—better than masking in every Trap A, by +0.118 (held-out ovary) to +0.330 (thyroid, from 0.400 to 0.730), and on every set of conflicting pairs—without the cost: it is never worse on a Trap B test and is within the registered margin on four of the six unaltered test sets. It met 40 of 42 components (Fig. 6); the two it failed are the thyroid split, where non-inferiority is inconclusive (-0.002 $[-0.011, +0.007]$), and ISIC 2020 overall (-0.021 $[-0.027, -0.016]$), where in-lesion hair still points to melanoma for most patients.

Nothing does better. A documented search over twelve further ideas on development data only—among them backdoor adjustment, product of experts, logit adjustment, moment matching, conditional adversarial debiasing, contrastive and invariance methods—and a second registered round of its three best on new data gave the same answer: IRMv1 met 41 of 42 components and lost only on ISIC 2020 overall (-0.009),

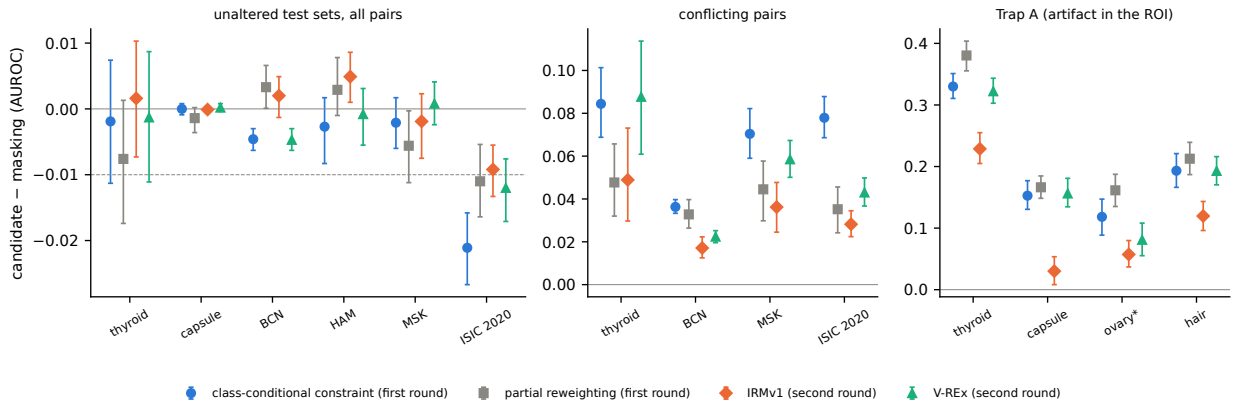


Figure 6: The four candidates closest to dominating masking, each minus masking (AUROC, crossed 95% CIs), on the confirmation data of their registered round (first round: class-conditional constraint, partial reweighting; second round, on new resamples: IRMv1, V-REx). Left: all pairs of the unaltered test sets (dashed: the registered margin −0.01). Middle: shortcut-conflicting pairs. Right: Trap A, the worse of reversed and correlated AUROC; *ovary held out from every choice.

and the constraint, refitted with its frozen recipe, replicated its result (loss on ISIC 2020 only, −0.015; Supplement S17). Every alternative to masking, including the unmasked model, loses overall on ISIC 2020, mostly on pairs whose images share the artifact status. No remedy we tested dominates masking.

Without artifact labels. Erasing from the masked features the subspace along which *generic* synthetic overlays move them (mask-then-erase, U-MtE; Supplement S8) removes much of what masking leaves for overlay-like artifacts (thyroid calipers in Trap A: from 0.400 to 0.624) at almost no cost on the unaltered test sets, but it is worse than masking for debris that resembles the pathology (capsule: 0.548 versus 0.667) and gains little where the shortcut is carried by correlates of hair (Table 4).

Decision guide. (i) Measure where artifacts sit relative to the ROI; for artifacts outside it, masking is enough. (ii) For artifacts inside it, with image-level artifact labels from an automatic detector, use the class-conditional constraint on the masked features, and expect a small overall cost where the deployment data still reward the shortcut; balancing gains more on the conflicting cases but costs more on the unaltered distribution. (iii) Without artifact labels, U-MtE helps for overlay-like artifacts; correlate-carried shortcuts need labels. (iv) Evaluate any remedy on shortcut-conflicting cases *and* on the unaltered distribution, with clean, correlated and reversed performance together, because a reversed-environment gain can be manufactured by removing diagnosis signal.

11. Discussion and limitations

What the evidence supports. Masking is not a neutral preprocessing step: its benefit depends on where the artifact sits. Four designs agree—real traps, covariate-matched traps, the transplant on fixed images and

controlled sweeps—and the effect holds for every masking implementation we tried, falls continuously with the real overlap, replicates on new patients and grows when images with contradicted labels are removed. The claim is about the *gap*: whether in-ROI masking helps a little or harms depends on whether the mask also removes disease context, which the linear-Gaussian account makes explicit and which differs between modalities. The transplant's paste-edge control shows that what matters is the location of *any* label-correlated pattern, not a particular artifact's appearance, so the problem cannot be solved by recognising particular artifacts; the evidence that real artifacts behave this way in natural images comes from the real and matched traps and the unaltered test data.

Why this is not obvious. That a mask cannot remove what it keeps sounds self-evident. Three findings go beyond it. Masking makes the model rely on an in-ROI artifact *more*, about twice as much in the ruler sweep, because it removes the competing evidence. The consequence is invisible in overall AUROC yet clinical: on the patient-disjoint thyroid split a fine-tuned model near the published benchmark misses more of the malignant nodules that carry a caliper, and part of that loss survives re-tuning the threshold on the test set itself. And the effect is predictable: fitted only to clean and correlated performance, a two-parameter account predicted 84 registered results before they existed. An ROI-masked model can therefore look unchanged, even after local recalibration, while failing on exactly the cases in which the shortcut points the wrong way; audits of masked models should stratify performance by artifact presence and location, and report reversed, correlated and clean performance together, because a reversed-environment gain can also be manufactured by removing diagnosis signal.

Limitations. (i) *Artifact labels.* Calipers and capsule debris are segmented automatically and hair masks are published semi-automatic masks; no clinician audited them. Independent sources confirm where hair lies but show that our hair-presence threshold over-calls hair; the thyroid caliper labels agree with two independent labellers where those agree with each other, but the registered cleaning rule could not be tested, and the ovary and capsule labels remain unchecked. A blinded rating by the first author, a non-clinician, is registered and incomplete. Removing contradicted images or using stricter definitions strengthens rather than weakens the results, and the transplant does not depend on the labels because the pasted instance is identical at both locations. (ii) *Patient identifiers and external validation.* TN3K/TNCD, MMOTU and SEE-AI publish no patient identifiers for the pooled images, so trap splits group near-duplicates by perceptual hash (the official thyroid test split is patient-disjoint). External validation on new patients was possible for dermoscopy only (Supplement S3). (iii) *Constructed shifts.* Reversed environments are constructed by subsampling; real shifts are less extreme, which is why we report the unaltered test data separately. (iv) *Frozen encoders.* Most analyses use frozen encoders with linear heads, below the best published fine-tuned models on the one cohort with a comparable benchmark (0.735 versus 0.773–0.799 AUROC). A fine-tuned model near that benchmark shows every registered harm, and the law replicates with fine-tuning in the three cohorts where

the network learned the task; the fine-tuned ovary network did not. (v) *Scope.* Chest radiography is a boundary case, and the linear-Gaussian account cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose. (vi) *Registration and statistics.* Commit times are set by the committing machine, so only registrations pushed to GitHub before their runs carry an external time stamp; the 16-test primary family was assembled after the individual results, and two sensitivity analyses were registered after the result they examine. Matching left within-label standardised differences of up to 0.19 for some covariates. (vii) *Thresholds.* Most of the thyroid sensitivity loss at validation-fixed thresholds is threshold transport; the part that survives re-tuning has a wide interval. (viii) *Remedies.* No remedy dominates masking; the best candidates fail on one cell, ISIC 2020 overall, by 0.01–0.02 AUROC, and the dominance tests use frozen DINOv2 features.

12. Conclusion

ROI masking is not a neutral preprocessing step. It removes shortcuts that live outside the ROI and leaves—or amplifies—those that live inside it; moving the same real artifact across the ROI boundary of the same images reproduces the effect, a two-parameter account predicts it, and it holds across masking implementations, encoders and new patients. On unaltered clinical data the cost falls on the cases in which the in-ROI artifact points to the wrong diagnosis: hidden in the overall AUROC, visible on those cases, and for a fine-tuned thyroid model partly beyond the reach of threshold re-tuning. Audits of masked models should report artifact location relative to the ROI and performance on shortcut-conflicting cases; where artifacts may lie inside the ROI and image-level artifact labels exist, a class-conditional constraint on the masked features should be used instead of masking alone.

References

[1] Bassi, P.R.A.S., Dertkigil, S.S.J., Cavalli, A., 2024. Improving deep neural network generalization and robustness to background bias via layer-wise relevance propagation optimization. Nature Communications 15, 291. doi:10.1038/s41467-023-44371-z.

[2] Belrose, N., Schneider-Joseph, D., Ravfogel, S., Cotterell, R., Raff, E., Biderman, S., 2023. LEACE: Perfect linear concept erasure in closed form, in: NeurIPS.

[3] Bissoto, A., Barata, C., Valle, E., Avila, S., 2023. Test-time selection for robust skin lesion analysis, in: MICCAI Workshops (ISIC).

[4] Bissoto, A., Valle, E., Avila, S., 2020. Debiasing skin lesion datasets and models? not so fast, in: CVPR Workshops, pp. 740–741.

[5] Brown, A., Tomasev, N., Freyberg, J., Liu, Y., Karthikesalingam, A., Schrouff, J., 2023. Detecting shortcut learning for fair medical AI using shortcut testing. Nature Communications 14, 4314. doi:10.1038/s41467-023-39902-7.

[6] Chuang, C.Y., Jampani, V., Li, Y., Torralba, A., Jegelka, S., 2023. Debiasing vision-language models via biased prompts. arXiv preprint arXiv:2302.00070 .

[7] Cohen, J.P., Viviano, J.D., Bertin, P., et al., 2022. TorchXRayVision: A library of chest x-ray datasets and models. MIDL .

[8] DeGrave, A.J., Janizek, J.D., Lee, S.I., 2021. AI for radiographic COVID-19 detection selects shortcuts over signal. Nature Machine Intelligence 3, 610–619.

[9] Fong, R., Vedaldi, A., 2019. Occlusions for effective data augmentation in image classification, in: ICCV Workshops.

[10] Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A., 2020. Shortcut learning in deep neural networks. Nature Machine Intelligence 2, 665–673.

[11] Germani, E., Zeineddine, F., Mourad, C., Albarqouni, S., 2026. When multi-institutional dataset aggregation masks shortcut-like behavior in foundation models for medical imaging. Medical Image Analysis 114, 104258. doi:10.1016/j.media.2026.104258.

[12] Gong, H., Chen, G., Wang, R., Xie, X., Mao, M., Yu, Y., Chen, F., Li, G., 2021. Multi-task learning for thyroid nodule segmentation with thyroid region prior, in: IEEE International Symposium on Biomedical Imaging (ISBI). doi:10.1109/ISBI48211.2021.9434087.

[13] Gong, H., Cheng, H., Xie, Y., Tan, S., Chen, G., Chen, F., Li, G., 2022. Less is more: Adaptive curriculum learning for thyroid nodule diagnosis, in: Medical Image Computing and Computer Assisted Intervention (MICCAI). doi:10.1007/978-3-031-16440-8_24.

[14] Hermann, K.L., Mobahi, H., Fel, T., Mozer, M.C., 2024. On the foundations of shortcut learning, in: International Conference on Learning Representations (ICLR).

[15] Holstege, F., Ravfogel, S., Wouters, B., 2025. Preserving task-relevant information under linear concept removal, in: Advances in Neural Information Processing Systems (NeurIPS). ArXiv:2506.10703.

[16] Jiménez-Sánchez, A., Juodelyte, D., Chamberlain, B., Cheplygina, V., 2023. Detecting shortcuts in medical images – a case study in chest x-rays, in: IEEE ISBI.

[17] Jin, Q., Gerych, W., Ghassemi, M., 2024. MaskMedPaint: Masked medical image inpainting with diffusion models for mitigation of spurious correlations. arXiv:2411.10686.

[18] Kina, M.A., Petersen, E., 2026. Prevalence calibration as shortcut mitigation. arXiv:2609.07922.

[19] Kirichenko, P., Izmailov, P., Wilson, A.G., 2023. Last layer re-training is sufficient for robustness to spurious correlations, in: ICLR.

[20] Li, Z., Evtimov, I., Gordo, A., Hazirbas, C., Hassner, T., Ferrer, C.C., Xu, C., Ibrahim, M., 2023. A Whac-A-Mole dilemma: shortcuts come in multiples where mitigating one amplifies others, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 20071–20082.

[21] Lin, M., et al., 2024. Shortcut learning in medical image segmentation, in: Medical Image Computing and Computer Assisted Intervention (MICCAI). doi:10.1007/978-3-031-72111-3_59.

[22] Makar, M., Packer, B., Moldovan, D., Blalock, D., Halpern, Y., D'Amour, A., 2022. Causally motivated shortcut removal using auxiliary labels, in: International Conference on Artificial Intelligence and Statistics (AISTATS), pp. 739–766.

[23] Maron, R.C., Hekler, A., Krieghoff-Henning, E., Schmitt, M., Schlager, J.G., Utikal, J.S., Brinker, T.J., 2021. Reducing the impact of confounding factors on skin cancer classification via image segmentation: technical model study. Journal of Medical Internet Research 23, e21695. doi:10.2196/21695.

[24] Nauta, M., Walsh, R., Dubowski, A., Seifert, C., 2022. Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis. Diagnostics 12, 40.

[25] Oakden-Rayner, L., Dunnmon, J., Carneiro, G., Ré, C., 2020. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging, in: ACM CHIL, pp. 151–159.

[26] Pedersen, N., Sydendal, R., Cheplygina, V., Sourget, T., 2026. Look what the probes dragged in! real-world chest x-ray shortcuts in MedCLIP. arXiv:2608.12086.

[27] Pewton, S.W., Cassidy, B., Kendrick, C., Yap, M.H., 2024. Dermoscopic dark corner artifacts removal: Friend or foe? Computer Methods and Programs in Biomedicine 244, 107986. doi:10.1016/j.cmpb.2023.107986.

[28] Ravfogel, S., Elazar, Y., Gonen, H., Twiton, M., Goldberg, Y., 2020. Null it out: Guarding protected attributes by iterative nullspace projection, in: ACL.

[29] Rueckel, J., Huemmer, C., Fieselmann, A., et al., 2021. Pneumothorax detection in chest radiographs: optimizing artificial intelligence system for accuracy and confounding bias reduction using in-image annotations in algorithm training. European Radiology 31, 7888–7900. doi:10.1007/s00330-021-07833-w.

[30] Sagawa, S., Koh, P.W., Hashimoto, T.B., Liang, P., 2020. Distributionally robust neural networks for group shifts, in: ICLR.

[31] Sourget, T., Hestbek-Møller, M., Jiménez-Sánchez, A., Xu, J.J., Cheplygina, V., 2025. Mask of truth: model sensitivity to unexpected regions of medical images. Journal of Imaging Informatics in Medicine doi:10.1007/s10278-025-01531-5.

[32] Stanley, E.A.M., Souza, R., Winder, A., Gulve, V., Amador, K., Wilms, M., Forkert, N.D., 2024. Towards objective and systematic evaluation of bias in artificial intelligence for medical imaging. Journal of the American Medical Informatics Association 31, 2613–2621. doi:10.1093/jamia/ocae165.

[33] Tschandl, P., Rosendahl, C., Kittler, H., 2018. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. Scientific Data 5, 180161.

[34] Veitch, V., D'Amour, A., Yadlowsky, S., Eisenstein, J., 2021. Counterfactual invariance to spurious correlations: Why and how to pass stress tests, in: Advances in Neural Information Processing Systems (NeurIPS). ArXiv:2106.00545.

[35] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., Summers, R.M., 2017. ChestX-ray8: Hospital-scale chest x-ray database and benchmarks, in: CVPR.

[36] Wang, Z., Xue, Y., Xi, H., Tan, X., Lin, K., Wang, C., Zhang, J., 2024. Influence of hair presence on dermoscopic image analysis by AI in skin lesion diagnosis. Computers in Biology and Medicine 183, 109335. doi:10.1016/j.compbiomed.2024.109335.

[37] Wegley, V., et al., 2026. Dermatology skin lesion image analysis: artifact annotation masks for ISIC 2019. Scholars' Mine, Missouri S&T, doi:10.71674/man1-qa33.

[38] Winkler, J.K., Fink, C., Toberer, F., et al., 2019. Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition. JAMA Dermatology 155, 1135–1141.

[39] Winkler, J.K., Sies, K., Fink, C., et al., 2021. Association between different scale bars in dermoscopic images and diagnostic performance of a market-approved deep learning convolutional neural network for melanoma recognition. European Journal of Cancer 145, 146–154.

[40] Yokote, A., et al., 2023. The SEE-AI project dataset: capsule endoscopy images with lesion bounding boxes. Kaggle dataset capsuleyolo/kyucapsule, CC BY 4.0.

[41] Zech, J.R., Badgeley, M.A., Liu, M., Costa, A.B., Titano, J.J., Oermann, E.K., 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLOS Medicine 15, e1002683. doi:10.1371/journal.pmed.1002683.

[42] Zhao, Q., Lyu, S., Bai, W., Cai, L., Liu, B., Wu, M., Sang, X., Yang, M., Chen, L., 2022. A multi-modality ovarian tumor ultrasound image dataset for unsupervised cross-domain semantic segmentation. arXiv preprint arXiv:2207.06799 .

[43] Zhong, Z., Zheng, L., Kang, G., Li, S., Yang, Y., 2020. Random erasing data augmentation, in: AAAI.